

Electronics and Telecommunications Engineering: (Machine Learning Project)

Problem Statement No. 40 – Network Intrusion Detection

The Challenge:

Create a robust network intrusion detection system (NIDS) using machine learning. The system should be capable of analyzing network traffic data to identify and classify various types of cyber-attacks (e.g., DoS, Probe, R2L, U2R) and distinguish them from normal network activity. The goal is to build a model that can effectively secure communication networks by providing an early warning of malicious activities.

Kaggle dataset link – <https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection>

Technology – Use of IBM Cloud Lite services is mandatory.

Scoping Decision and Implementation

The dataset explicitly linked above (Kaggle: sampadab17/network-intrusion-detection, derived from NSL-KDD) contains binary labels — normal and anomaly — rather than the individual attack categories (DoS, Probe, R2L, U2R) described in the challenge. Since the linked dataset does not contain the original per-attack-type labels, this project implements binary anomaly detection, consistent with the dataset actually provided. This is a deliberate, documented scoping decision made after verifying the dataset directly, since the explicitly linked dataset takes precedence over the prose description when the two conflict.

The system is built in two layers. The cloud layer uses IBM Watson Studio AutoAI to automatically preprocess data, select an algorithm, and tune hyperparameters, producing a trained classification model that is deployed to Watson Machine Learning as a live REST scoring endpoint. The local layer captures raw network packets using Scapy, reconstructs the relevant NSL-KDD-aligned connection features through a custom flow aggregator, and sends each completed connection to the deployed model for real-time classification.

Each connection flagged as anomalous is further passed through a rule-based heuristic layer that estimates the likely attack pattern (e.g. scanning-like or flood-like behavior) based on known feature signatures, and paired with a recommended mitigation action for the operator. Results are displayed on a live Streamlit dashboard showing the connection feed, active alerts, and mitigation guidance, giving operators an early warning of potential malicious activity as it happens rather than relying on delayed manual review of static logs.